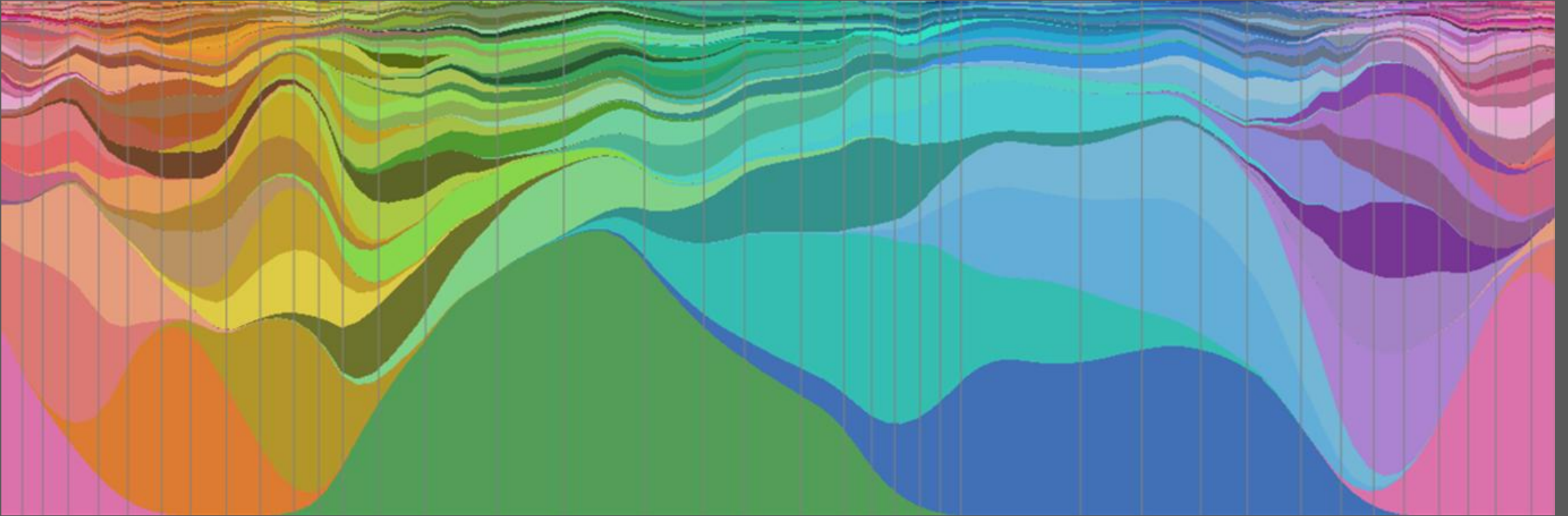


LDSCI5209 - Information Presentation and Visualisation

Week 5: Interaction and Animation



Dimitris Mylonas

Northeastern University London

Recap: from cones to words

- Colour perception
- Colour cognition
- Colour dimensions and colour spaces
- Colour encoding and colour palettes

Plan for today

- Personas in interaction design
- Interaction
- Animation

Personas

User needs, objectives, challenges, technologies

What are personas?

- Personas are fictional characters based on data, representing various user types.
- They help tailor visualisation designs to specific audiences.
- Personas promote effective communication by simulating user interactions.
- They ensure visualisations meet user goals and expectations, enabling informed design decisions.

<https://www.interaction-design.org/literature/article/personas-why-and-how-you-should-use-them>

Creating Personas

- Gather data through research, interviews, and surveys to understand user needs.
- Identify key user segments based on demographics, goals, and expertise.
- Create detailed personas with names, backgrounds, and motivations for each segment.
- Continuously update personas based on feedback and insights to align with design goals.



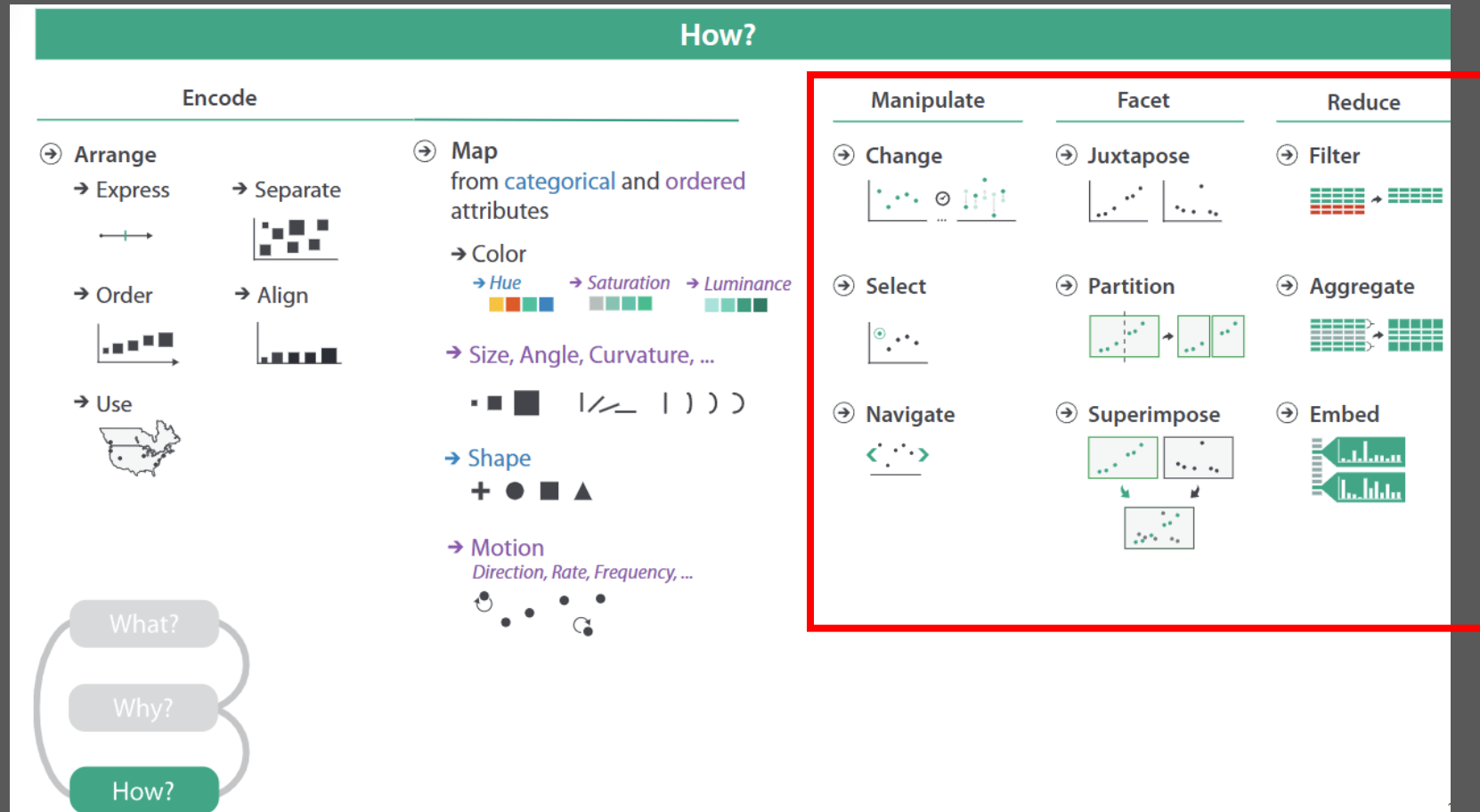
Interaction

Selection, changes over time, navigation, brushing and linking, query

Why interaction in data visualisation?

- Reduces complexity
- Enables data exploration
- Makes it personal to the viewer
- Allows deeper dives

How to enable interaction



*“Overview first, zoom and filter, and
details on demand.”*

Ben Shneiderman “The Shneiderman Mantra”

Key concepts of interaction

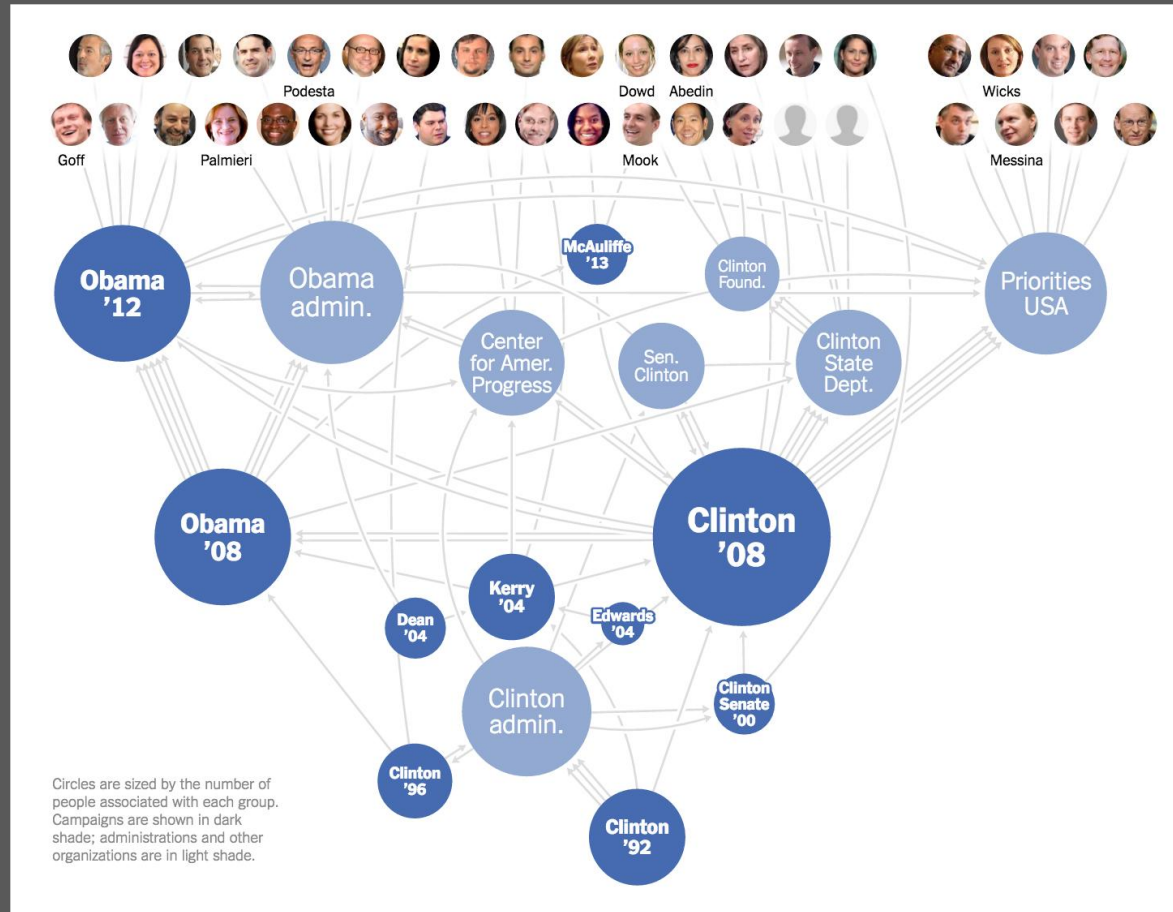
- **Change over time:** Encodings, Animated Transition
- **Selection:** Highlight
- **Navigation:** Pan/Translate, Rotate, Zoom

Change over time



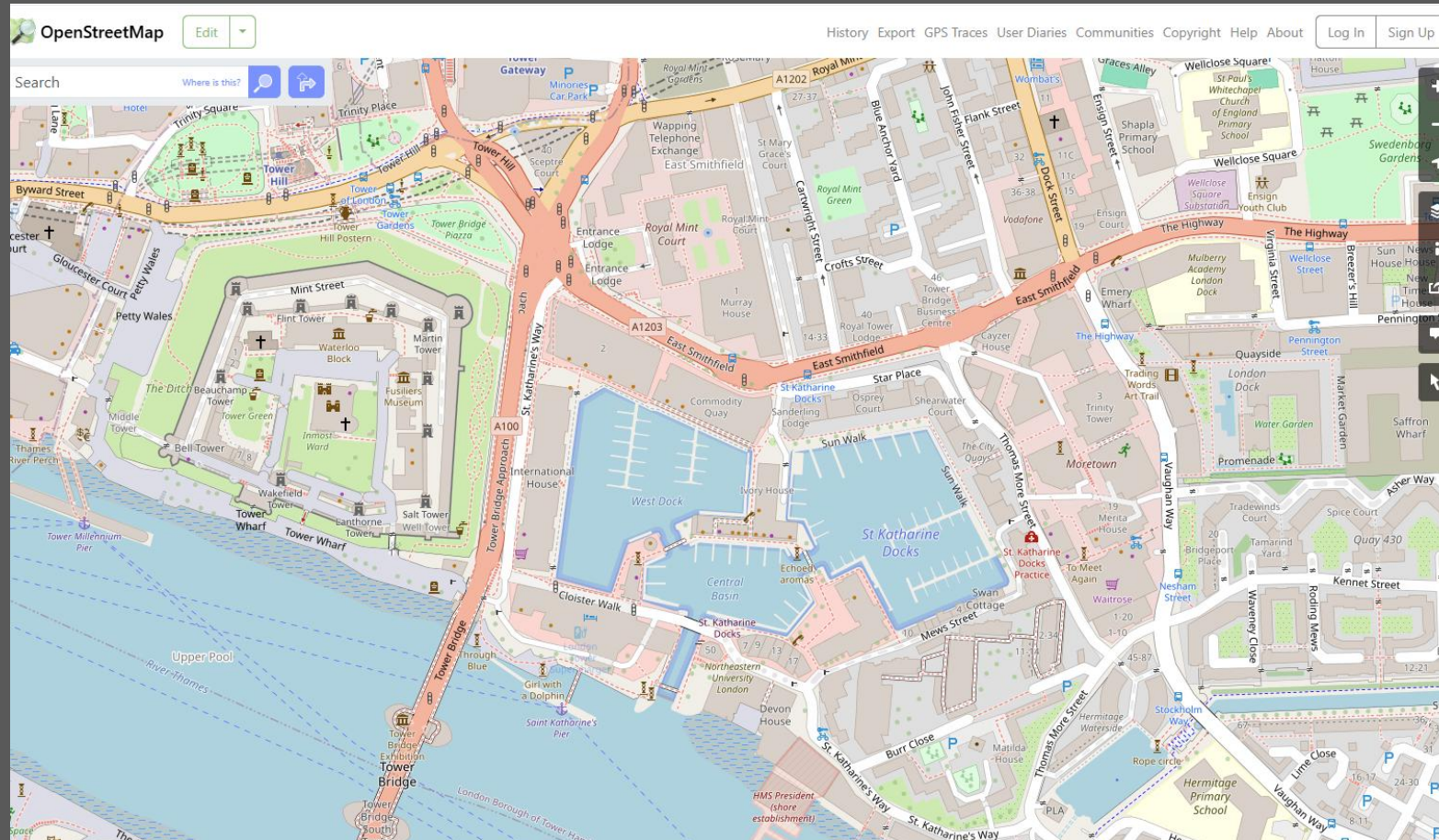
[https://www.gapminder.org/tools/#\\$chart-type=bubbles&url=v2](https://www.gapminder.org/tools/#$chart-type=bubbles&url=v2)

Selection



<https://www.nytimes.com/interactive/2015/05/17/us/elections/2016-presidential-campaigns-staff-connections-clinton-bush-cruz-paul-rubio-walker.html>

Navigation

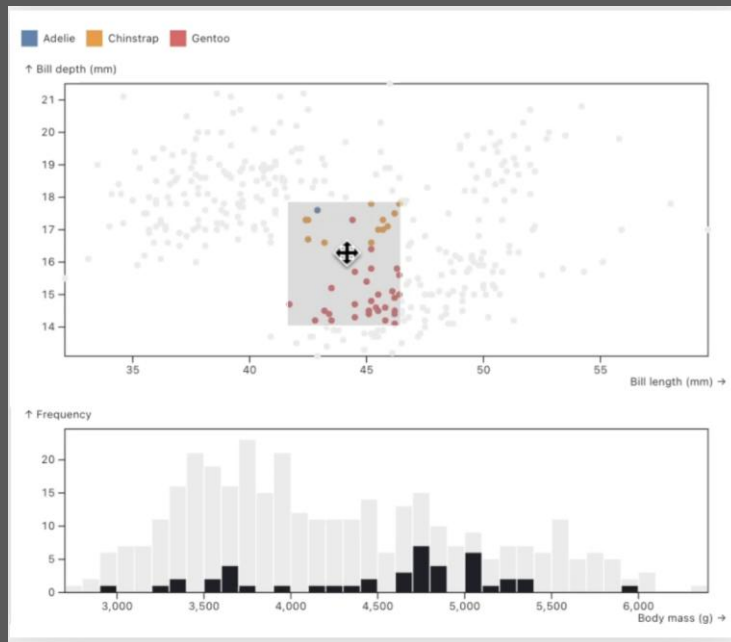


<https://www.openstreetmap.org/#map=17/51.507464/-0.071647>

Brushing and linking

Brushing: direct attention to a subset of data

Linking: see selected data in other views



<https://observablehq.com/blog/linked-brushing>

Queries and filtering

Manipulate

- ➔ **Change over Time**
- ➔ **Select**
- ➔ **Navigate**
 - ➔ **Item Reduction**
 - ➔ **Zoom**
Geometric or Semantic
 - ➔ **Pan/Translate**
 - ➔ **Constrained**
 - ➔ **Attribute Reduction**
 - ➔ **Slice**
 - ➔ **Cut**
 - ➔ **Project**

Rome2Rio [My trips](#) [Hotels](#) [Cars](#) [Sign in](#)

London, England ↔ Boston, MA, USA [Add destination](#)

4 ways to travel to Boston

Fly from Heathrow Airport	BEST
✈️ 10h 15min	£225-804 >
Train to London Gatwick Airport, fly	CHEAPEST
🚆 ✈️ 10h 37min	£171-643 >
Fly from London City Airport	
✈️ 12h 24min	£323-436 >
Fly from London Stansted Airport	
✈️ 12h 42min	£286-399 >

[Add destination](#)

Hotels in Boston >

<https://www.rome2rio.com/map/London/Boston#trips>

Animation

Motion, transitions, dynamics, storytelling

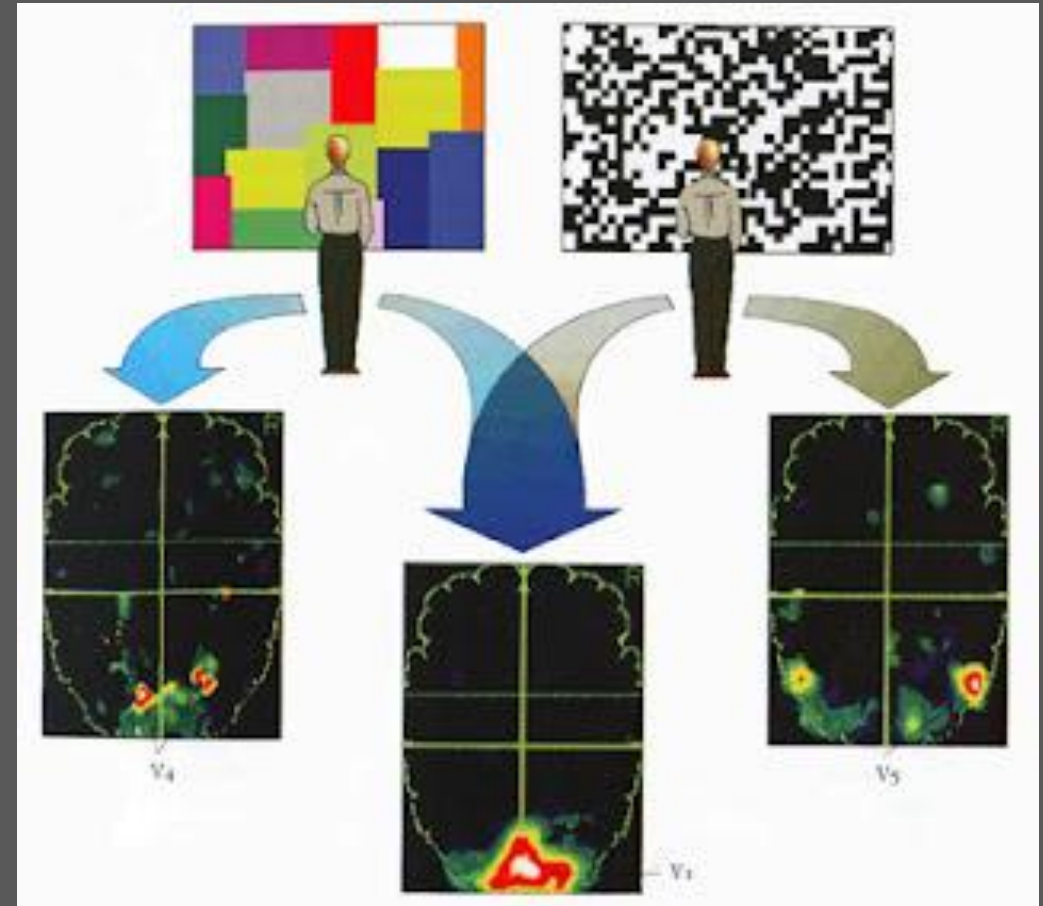
Why animation in data visualisation?

- Visual variable to encode data
- Direct attention
- Understand system dynamics
- Understand state transition
- Increase engagement



Motion perception

- Motion is perceived at about ~10 frames/sec (100 ms).
- Pre-attentive, stronger than colour or shape
- More sensitive to motion at periphery
- Similar motions perceived as a group

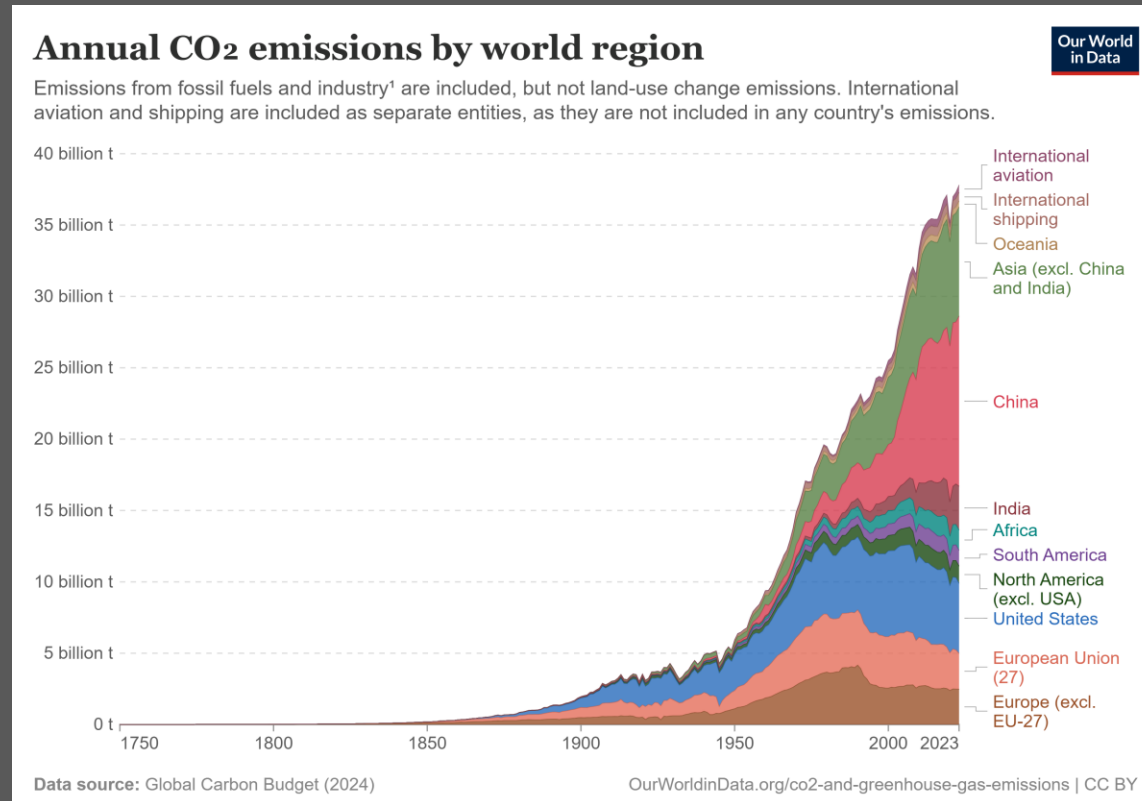


Brain activations for visual stimuli, Zeki (1993)

Advantages and disadvantages

Pros	Cons
Engages the audience	Cognitive overload
Reveals changes over time	Slower information retrieval
Enhances storytelling	Technical complexity
Highlights key insights	Accessibility concerns

Interactive animations



<https://ourworldindata.org/co2-emissions>

Questions?